

BREEDING AND FERTILITY OF THE EGYPTIAN SPINY MOUSE, *ACOMYS CAHIRINUS*: EFFECT OF DIFFERENT ENVIRONMENTS

by

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SUMMARY

2 years experience with *Acomys* in one laboratory confirmed the slow breeding rate found by other workers. On moving the animals to another laboratory there was a 2-5 fold increase in breeding rate. The most likely explanation of this change was the presence of copper in the drinking water at the 1st laboratory, due to the use of copper water pipes. The data relating to maintenance and breeding of *acomys* for 2 years at the 2nd laboratory were analyzed to provide a more representative picture of the species.

The Egyptian spiny mouse, *Acomys cahirinus*, is a native of the sandy deserts and rocky terrain of the Middle East. It belongs to the genus *Acomys* of the same subfamily, Murinae, that contains common rats and mice (Ellerman, 1941, 1949). Although first described in 1819, it has only recently become the subject of laboratory investigation, and is now being increasingly used for metabolic, genetic and parasitic studies. This is shown by the number of published papers relating to this species: 1 in 1925-49; 9 in 1950-59; 33 in 1960-69; 38 in 1970-73. The principal feature of interest in the spiny mouse is the diabetes, obesity and hyperplasia of the islets of Langerhans seen in a colony of *A. cahirinus* kept in Geneva (Gonet, Stauffacher, Pictet & Renold, 1965); this colony was derived from one kept in Basel (Flückiger & Operschall, 1962; Hefti & Flückiger, 1967). Gonet *et al.* (1965) and Strasser (1968) have given details of the maintenance and breeding of this species. In particular, it has been emphasized that the relatively long gestation period (39 days), small litter size (1-4, generally 2) and 'reproductive idiosyncrasies' impose a severe limitation on the rate of development of a laboratory colony. Although this was confirmed by my earlier experience with 3 colonies (from 3 different

sources, including the Basel-Geneva colony), a remarkable increase in reproductive performance was subsequently found on moving to another laboratory.

MATERIALS AND METHODS

Animals

The 3 colonies considered in this paper were derived from colonies that were founded with animals obtained from Israel; they were maintained at the Nuffield Institute of Comparative Medicine in London 1969-71, and subsequently at the Wellcome Research Laboratories in Beckenham.

Colony G. Animals from the Basel-Geneva colony, established temporarily at Guy's Hospital Medical School 1966-68 (Wise, 1968).

Colony J. Breeding animals from the colony at the Hebrew University-Hadassah Medical School, Jerusalem, a gift from Professors E. Shaffrir and Y. Gutman.

Colony Z. Breeding animals from the colony exhibited in the Garden of the Zoological Society of London.

Lighting, temperature and diet

All the *Acomys* were kept in natural daylight supplemented with artificial lighting (London, incandescent; Beckenham, fluorescent) of constant duration (14 h). The temperature of the animal rooms was kept above 20°C at both establishments. Pelleted diet (MRC 2; Scientific Products Farm, Ash, Canterbury, Kent—later Charles River U.K., Ltd., Manston Road, Margate, CT9 4LT), placed in hoppers, and tap water were supplied *ad libitum*.

Caging

The most convenient arrangement for breeding was 1 male and 2 breeding females to a cage, as was found by Gonet *et al.* (1965) and Strasser (1968). The animals were kept in polypropylene cages of various sizes, the most suitable being mouse boxes (48 × 15 × 13 cm) with solid bottoms and stainless-steel tops (North Kent Plastic Cages Ltd, Home Gardens, Dartford, Kent). However at the Nuffield Institute where fertility was suboptimal, better breeding results were obtained with Colony Z, but not with the other colonies, by allowing 2 or more breeding generations to remain in a colony box; this resembled closely the conditions under which the colony was maintained in the zoological gardens.

Fostering of young to other lactating *Acomys* was uniformly successful, at ages as early as 2 days. A precaution, probably essential, to prevent rejection is to warm the young in the hand before transfer. The usual weaning age is 28 days, but when necessary *Acomys* were weaned as early as 17 days; brown

bread and pellets both crumbled in milk, were given, together with small fragments of apples and carrots.

Water supplies

The water supply at the Nuffield Institute was mains water of the Metropolitan Water Board. It is river water in origin and has moderate to high total hardness (240 mg/l; Ca^{2+} , Mg^{2+} , HCO_3^- & SO_4^{2-}). The water pipes in the Institute were of copper. The water supply at the Wellcome Laboratories was well water from deep bore holes in the chalk. It was characterized by high contents of calcium bicarbonate and free carbon dioxide and had a total hardness (240 mg/l) comparable with that of water at the Nuffield Institute. The water pipes were of galvanized iron and consequently zinc levels could be high especially after overnight stasis. Samples of water from both establishments were analyzed for iron, lead, chromium, manganese, copper and zinc. Copper and zinc were determined by atomic absorption spectrophotometry using a Pye-Unicam SP1900 (Pye-Unicam Ltd, York Street, Cambridge) with an air/acetylene flame. Standards were prepared according to the appropriate manual (Perkin-Elmer Corporation, 1973). Sensitivity ($\mu\text{g/ml}$) and detection limit ($\mu\text{g/ml}$) were for copper 0.026 and 0.002, and for zinc 0.009 and 0.002, respectively. There were no differences between the water supplies except for copper and zinc, the data for these are given in Table 1. The pellet diet also was analyzed for zinc; the value obtained was 50.4 mg/kg, which is the same as the manufacturer's specification.

Table 1. Concentration of zinc and copper ions in samples of tapwater from the *Acomys* rooms. Concentrations ($\mu\text{g/l}$) are given as mean $\pm$ s.e. Number of samples, taken on different days over a period of months, in parentheses.

	<i>Zinc</i>	<i>Copper</i>
Nuffield Institute	undetected (6)	155.7 ± 44.0 (7) range 30-305
Wellcome Laboratories		
room 1 (1st year)	21.0 ± 3.1 (3) range 15-25	15.2 ± 6.3 (6) range 0-45
room 2 (2nd year)	950 ± 540 (4) range 100-2460	11.5 ± 4.6 (4) range 0-20

RESULTS

Breeding records at the Wellcome Laboratories

Litter size

There were significant differences amongst the 3 colonies (Table 2). For colonies G and J, χ^2 analysis based on classes of size 1, 2 and >2 showed that

Table 2. Number and size of litters of *Acomys cahirinus* born over a 2-year period at the Wellcome Laboratories.

Colony	Breeding groups females		Litter size							Total litters	Total young	Weaned young		
			1	2	3	4	5	6	7			♂	♀	total
G	24	39	15	45	45	11	0	0	0	116	284	115	113	228
J	16	25	24	46	28	5	0	0	0	103	220	101	94	195
Z	21	37	36	61	54	26	5	1	1	184	462	204	219	425
Total	61	101	75	152	127	42	5	1	1	408	966	420	426	846

litter size was significantly different, there being fewer singletons and more triplets for G ($\chi^2=7.289$, d.f. 2, $p<0.050>0.025$). For colonies G and Z, the most obvious difference was the greater number of litters in Z larger than 3. Analysis was accordingly based on classes of size 1, 2, 3 and >3 , and the difference was significant ($\chi^2=7.866$, d.f. 3, $p<0.050>0.025$). Comparison of colonies J and Z, with either of these sets of class intervals showed that litter size was significantly larger for Z ($\chi^2=6.46$, d.f. 2, $p<0.050>0.025$; $\chi^2=11.428$, d.f. 3, $p<0.010>0.005$), Z having many more litters of 4 or more than did J.

Sex ratio

The sex ratio at weaning and the data for the 3 colonies are given in Table 2. The overall sex ratio is 1:1, with no significant departures from this ratio for the individual colonies. The analysis is limited by a possible higher infant mortality of one sex, although this is unlikely since individual deaths were too small in number to affect the ratio significantly, and litter mortality for colony G was significantly higher than that for the other colonies without affecting the sex ratio for the colony.

Neonatal mortality

Almost all deaths before weaning occurred within the first 2 days of life. The mortality may be considered as loss of whole litters and individual deaths.

Loss of whole litters. Total neglect of a litter after birth leads to death within a few hours. The reason for rejection could be failure of lactation. The babies are rejected once the body temperature falls. The incidence of loss of whole litters (Table 3) was significantly higher in colony G than in the others ($\chi^2=21.05$, d.f. 1, $p<0.001$), where there was no difference ($\chi^2=0.017$, d.f. 1). For colony G, but not for J and Z, there was a significant increase in mortality of litters ($\chi^2=4.13$, d.f. 1, $p<0.050>0.025$) when there were 2 adult females per cage rather than 1. In fact when the data for 1 and 2 females for colony G were compared separately with the combined data for either J

Table 3. Neonatal mortality in litters of *Acomys cahirinus* over a 2-year period at the Wellcome Laboratories.*

Colony	Breeding groups		Number of litters born						% dead litters		
	1 ♀	2 ♀	alive			dead			1 ♀	2 ♀	total
G	9	15	42	58	100	11	34	45	21	37	31
J	7	9	34	63	97	2	13	15	6	17	13
Z	5	16	20	156	176	5	21	26	20	12	13
Totals	21	40	96	277	373	18	68	86	16	20	19

*The discrepancy between the sum of live and dead litters in this table and the total litters shown in Table 2 represents those dead litters in which the number of offspring born was not known.

or Z, it was found that the significantly higher mortality for colony G was entirely accounted for by the higher mortality in the 2-female group ($\chi^2=27.0$, d.f. 1, $p<<0.005$). Thus this analysis affords no support for the survival value of the 'midwife' *Acomys*; in fact for colony G, it appears to be better to restrict the breeding group to 1 male and 1 female per cage. There was a concentration of litter deaths in certain breeding groups (e.g. 70% of all dead litters in G occurred in only 7 out of the 24 breeding groups), and 3 successive dead litters was used as a criterion for rejecting breeders. However, omitting such rejected groups from the above analyses of litter deaths did not alter the results.

Individual deaths. There were a small number of such deaths, with no significant differences between colonies: G, 3 (1.3% of total); J, 4 (2.1%); and Z, 7 (1.7%).

Seasonal variation in fertility

Analysis of the breeding data for the 3 colonies over a 2-year period shows a well-marked seasonal variation in fertility (Table 4). The year was divided into 3 periods of 4 months starting with 1 February (the first month's data available, but also conveniently 1 gestation period after the winter solstice), and the data analyzed by the Friedman 2-way analysis of variance by rank (Siegel, 1956). Whether considered on the basis of litters born ($p=0.002$), litters weaned ($p=0.008$) or young weaned ($p=0.006$), the seasonal variation was significant. The data for Oct-Jan were significantly less than those for either of the other 2 periods ($p<0.03$, *Randomization test for matched pairs* in Siegel, 1956). Although the fertility appeared higher in Feb-May than in the summer period, the difference was not significant. These results agree with observations made of *Acomys cahirinus* in the wild, presumably in Egypt,

where births are mostly between Feb-Sept (Walker, 1964), but for *Acomys* in Israel breeding is said to be limited to the spring (Bedrak, Samoiloff & Finkelstein, 1971).

Table 4. Seasonal variation in number of litters of *Acomys cahirinus* born in 3 colonies at the Wellcome Laboratories.

Year	<i>Litters per breeding group per 4 months</i>					
	G		J		Z	
	1st	2nd	1st	2nd	1st	2nd
Feb-May	3.1	1.9	2.9	3.6	3.0	4.6
June-Sep	2.6	1.2	2.5	2.7	3.2	4.1
Oct-Jan	1.5	0.6	2.3	2.0	2.6	2.1

Bodyweight

For different groups of males of colony G, the mean bodyweight (g) $\pm$ s.e. at various ages were: 8 months 45.5 ± 1.5 (11); 12 months 49.4 ± 2.0 (9); 18 months 51.0 ± 0.9 (5); 24 months 54.0 ± 1.1 (5). Data for colony J were: 8 months 45 ± 1.2 (9); and for colony Z: 8 months 51.6 ± 1.0 (19), and 24 months 58.4 ± 1.1 (5). For older animals (3-4 years), the mean for colony G was 63.4 g; for J was 64.8 g, and for Z it was 71.1 g. The weight records at the Nuffield Institute were not significantly different from these: information was also collected for colony G at younger ages: 15 days 14.6 ± 0.6 (5); 30 days 24.2 ± 1.2 (5); 50 days 30.4 ± 1.1 (5); 72 days 32.6 ± 1.2 (5) and 3 months 34.5 ± 1.4 (5). These bodyweights are very different from those for the same strain (colony G) in Geneva (Gonet *et al.*, 1965; Pictet, Orzi, Gonet, Rouiller & Renold, 1967): 28 days 32 (range 30-33); 70 days 40 (35-75); and 9 months 70 (45-95). These wide ranges in weights were due to a variable obesity, which probably resulted from the addition of sunflower seeds to the diet (Hefti, 1971).

Breeding records at the Nuffield Institute

The most obvious difference between the breeding records at the two establishments was the much lower fertility at the Nuffield. Although this is evident from the number of *Acomys* born in 2-year periods in each laboratory, account must be taken of the difference in the breeding programmes. At Beckenham breeding was limited to the pairs initially selected and their replacements, whereas at the Nuffield Institute all mice born were included in the breeding programme. A satisfactory basis for comparison is the ratio of live births to breeding years (Table 5), and it is clear that all the colonies were very much more prolific at Beckenham. In addition, litter size was also affected. Thus of the 26 litters born in colony G at the Nuffield Institute,

Table 5. Reproductive performance of *Acomys cahirinus* at the Nuffield Institute and the Wellcome Laboratories.

Colonies	Nuffield Institute			Wellcome Laboratories		
	G	J	Z	G	J	Z
Initial breeders*						
♀	10	14	9	23	16	22
♂	9	13	1	17	12	14
Offspring (A)†	55 (42)	54 (46)	68 (60)	284 (228)	220 (195)	462 (425)
Breeding years (B)‡	18.5	30.3	31.2	35.7	21.4	38.3
Ratio A/B§	3.0 (2.3)	1.8 (1.5)	2.2 (1.9)	8.0 (6.4)	10.3 (9.1)	12.1 (11.1)

*At the Wellcome Laboratories breeding was limited to the pairs initially selected and their replacements: at the Nuffield Institute all mice born were included in the breeding programme. †Live births; number of young weaned in parentheses. ‡Total lifetime of females (age >3 months) kept with males. §Ratio for live births; that for weaned young in parentheses.

20 were twins (the remainder were 2 singles, 3 sets of triplets and 1 set of quadruplets), whereas at Beckenham only 45 out of 116 litters were twins (Table 2). The difference is significant ($\chi^2=12.44$, d.f. 1; $p<0.005$).

Overall neonatal mortality was slightly higher at the Nuffield Institute, although the numbers are too small for significance; furthermore, the colonies had the same ranking order. Thus mortality for colony G was 23.6% (Wellcome 19.7%), for J 17.4% (11.4%), and for Z 13.3% (8.0%).

DISCUSSION

On moving the animals from the Nuffield Institute to the Wellcome Laboratories both litter size and frequency increased. The latter is the more important since it largely accounts, at least for colony G, for the increased breeding rate. This is shown by comparison of the ratio of the numbers of litters at the two places ($116(W)/26(N)=4.5$) with the ratio of the number of offspring ($284(W)/55(N)=5.2$). If these changes are linked it would indicate that the breeding rate obtained at Beckenham was in excess of those obtained by Gonet *et al.* (1965) and by Strasser (1968); for those authors state that the litter size is predominantly 2, whereas at Beckenham for the same strain (colony G) only 40% of litters were twins, 50% being >2 (Table 2). In this respect it is interesting that the colony in Geneva was given sunflower seeds as a dietary supplement to improve reproduction (Cameron, Stauffacher, Orci, Amherdt & Renold, 1972).

There were at least 4 differences in the conditions of the *Acomys* at the 2 laboratories: animal attendants, lighting, the presence of other species, and drinking water. The first can be discounted, for although the staff at the Wellcome Laboratories were more experienced, the attention given to the

Acomys at the Nuffield Institute was excellent. Fluorescent lighting does have an emission in the ultraviolet in contrast to incandescent lamps, but since the animal room at the Nuffield Institute had a large window on the outside of the building, this seems an unlikely factor. The presence of other species (rats, hamsters, guinea-pigs, gerbils and steppe lemmings at various times) throughout the 2 years at the Nuffield Institute is a likely explanation, only countered by the satisfactory breeding records of a few *Acomys* placed in a general animal room at the Wellcome Laboratories. Happold (1969) has reported for the northern Sudan that where there are many other species of small mammals *Acomys* are not common, whereas in the semi-desert jebels *Acomys* are abundant to the exclusion of other species. He does not think this is due to competition and exclusion of the other species; such distribution therefore supports the possibility that *Acomys* is sensitive to the presence of other animal species.

The last and most likely explanation of the change in reproductive performance is the presence of copper in the water supply of the Nuffield Institute (Table 1), as copper is a competitor of zinc at the tissue level (Evans, 1973), and zinc is essential for reproduction. Thus dietary zinc deficiency in rats causes testicular atrophy, decreased size of accessory sex organs and partial to complete atrophy of the germinal epithelium (Millar, Fisher, Elcoate & Mawson, 1958). This may be due to a failure of gonadotrophin secretion, although those authors note that since spermatozoa contain a very high concentration of zinc, it seems likely that zinc deficiency would cause failure of sperm production without interference with hormone production. In man, zinc deficiency likewise affects sexual development especially in the male, and the associated endocrine manifestations were considered compatible with hypopituitarism (Prasad, 1967). The presence in the diet of zinc in quantities much greater than those of copper in the water is not incompatible with this argument, for much of the zinc in food is not absorbed from the gut because of complexing with various substances, particularly phytic acid—inositol hexaphosphoric acid (Oberleas, Muhrer & O'Dell, 1966). The level of phytate depends on the plant content of the diet, in particular grasses and cereals. The zinc deficiency syndrome in man prevalent in Iran and Egypt is due to a diet mainly of wheat rich in phytate (Reinhold, Nasr, Lahimgarzadeh & Hedayati, 1973). The pellet diet of the *Acomys* was in fact high in cereal content (30% wheat, 20% oats, 10% each of barley, maize and soya bean meal). Furthermore, zinc deficiency syndrome occurs in laboratory animals fed soya bean protein (Davies, 1972). Similar considerations apply to the dietary absorption of copper by the gut (Evans, 1973), which will not apply to its absorption from drinking water. Thus it could be argued that for these *Acomys*, which are probably borderline as regards zinc intake, the absorption of small quantities of copper could reduce fertility. The copper present in the

drinking water is the result of its passage through copper pipes. The growing use of copper piping in animal houses of research institutes and zoos could therefore be an important limitation in the breeding of some species. Treatment of the water with an ion-exchange resin would resolve this problem.

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